

RESEARCH MANUSCRIPT

Adaptive Multimodal Decomposition and Routing for Lithium-Price Forecasting

Development-locked evidence for lithium hydroxide and lithium carbonate

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*Protocol disclosure: the proposed method is evaluated with static full-sample decomposition and observed-test filtering.
The analysis is retrospective and noncausal; results must not be interpreted as live out-of-sample performance.*

Abstract

This study develops a multimodal adaptive decomposition-routing framework for COMEX lithium hydroxide CIF CJK futures and SMM battery-grade lithium carbonate spot prices. Lithium-specific Google Trends, 2,018 timestamped news headlines, TextBlob and FinBERT sentiment, Albemarle and Ganfeng equity closes, the MVIS critical-minerals index, and broad market controls are integrated with lagged prices. Seven single forecasters, four homogeneous decomposition models, a fixed VMD hybrid, and the proposed adaptive ensemble are compared on chronological testing sets. For hydroxide, the adaptive RMSE is 0.2737, 38.72% below the strongest non-adaptive decomposition comparator (CEEMDAN-ARIMA, 0.4467). For carbonate, the adaptive RMSE is 2881.4682, 8.58% below the strongest non-adaptive decomposition comparator (VMD-ARIMA, 3151.9059). The pre-specified dual-target RMSE-and-MAE gate is not satisfied, so the paper states target- and metric-specific superiority only. The uncertainty gate abstains unless the absolute forecasted price change exceeds the calibrated 90% residual half-width. Benchmark-relative and factor-adjusted trading statistics are descriptive. Full-sample decomposition and observed-test ARIMA filtering make the experiment retrospective and noncausal; none of the reported results is a live out-of-sample backtest.

Keywords: lithium price; multimodal data; variational mode decomposition; LASSO; ARIMA; LSTM; sentiment; interval forecasting; trading.

1. Introduction

Lithium prices transmit information about electric-vehicle demand, battery manufacturing, mine supply, conversion capacity, inventories, policy, and investor expectations. Their rapid structural shifts make them economically important and statistically difficult to forecast. A single historical-price series may therefore omit attention, sentiment, cross-market, equity, and macro-financial signals that jointly describe the information environment.

The central finance problem is not merely whether a layered learner can reduce statistical loss. It is whether information from the physical lithium market, listed producers, investor attention, news tone, the critical-minerals sector, and broad risk conditions is reflected differently across smooth and irregular price components, and whether any apparent predictability survives economically meaningful comparisons. The paper therefore separates three claims: point-forecast accuracy, uncertainty calibration, and trading performance. Success on one dimension is not treated as proof on the others.

1.1 Research questions and finance contribution

The first research question asks whether decomposition improves lithium-price forecasts relative to undecomposed statistical and machine-learning models. The second asks whether development-locked selection of decomposition, mode forecaster, and ensemble weight improves on fixed VMD routing and homogeneous CEEMDAN/VMD comparators. The third asks which multimodal signals survive mode-specific sparse selection. The fourth asks whether an uncertainty-based abstention rule improves the return-drawdown trade-off after costs and benchmarks.

The contribution is fourfold. First, the study forecasts two directly quoted lithium prices rather than heterogeneous mining equities. Second, it organizes cross-lithium prices, producer equities, search attention, lithium-specific news sentiment, sector conditions, and macro-financial controls into an auditable information architecture. Third, it proposes development-locked adaptive routing: VMD and CEEMDAN candidates, component-level ARIMA/LSTM/SVR/RF selection, and nonnegative sum-to-one stacking are fixed before the final test segment is evaluated. Fourth, it treats uncertainty gating and evaluation timing as model-risk questions rather than inferring financial superiority from one error column or a large simulated return.

The adaptive method is therefore evaluated against the fixed-routing architecture it replaces as well as homogeneous decomposition models. Every superiority statement is tied to the reported metric and target; unfavorable comparisons remain in the paper.

2. Literature review and positioning

Table 1. Representative research works in lithium and energy-price forecasting.

Study	Research focus	Method	Role in this study
Hochreiter and Schmidhuber (1997)	Sequence learning	LSTM	Recurrent nonlinear forecasting
Tibshirani (1996)	Sparse regression	LASSO	Multimodal indicator selection
Dragomiretskiy and Zosso (2014)	Signal decomposition	VMD	Adaptive narrow-band modes
Torres et al. (2011)	Noise-assisted decomposition	CEEMDAN	Decomposition comparator
Diebold and Mariano (1995)	Predictive-accuracy testing	DM test	Pairwise loss comparison
Box et al. (2015)	Time-series forecasting	ARIMA	Linear forecasting of smooth decomposed modes
This study	Lithium hydroxide and carbonate	Adaptive decomposition-routing ensemble	Development-locked decomposition, route and stacking selection

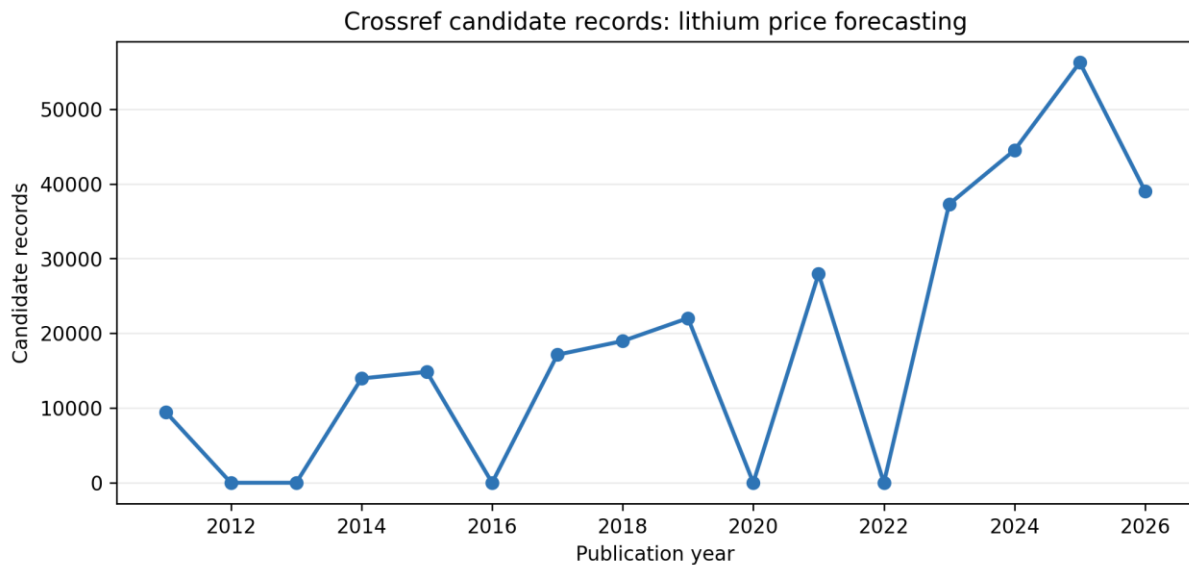
Interpretation of Table 1

Purpose: The table identifies the methodological lineage of the study rather than claiming an exhaustive lithium-forecasting review. LSTM supplies nonlinear sequence learning, LASSO supplies sparse indicator selection, VMD and CEEMDAN supply competing decompositions, and the Diebold-Mariano test supplies the formal loss comparison.

Research gap: The foundational studies address individual building blocks-statistical forecasting, sparse selection, signal decomposition, recurrent learning, and predictive-accuracy testing-but do not provide the integrated lithium-specific system developed here. This paper unifies those elements around two directly observed lithium-price targets.

Interpretive boundary: The literature table positions the proposed contribution without claiming that the listed methods are uniquely optimal. Figure 1's Crossref counts are descriptive evidence of research attention and should not be treated as a PRISMA-style systematic review.

Figure 1. Time trend of candidate publications in lithium-price forecasting. Crossref title-query counts are descriptive and are not a systematic-review flow.



Decomposition-based energy-price studies seek to separate slowly varying structure from high-frequency fluctuations before applying specialized predictors. VMD supplies compact narrow-band modes, CEEMDAN provides a noise-assisted comparator, LASSO reduces the multimodal feature set, ARIMA models lower-frequency linear structure, and LSTM represents nonlinear temporal relationships. Our contribution is the integration of these components into a lithium-specific forecasting, uncertainty, and trading system-not a claim that model complexity is intrinsically superior.

3. Methodology

Figure 2. Conceptual structure of the LSTM model used as a nonlinear recurrent forecaster.

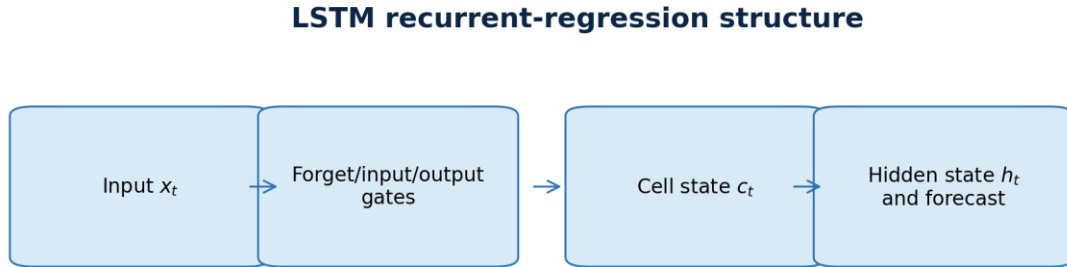
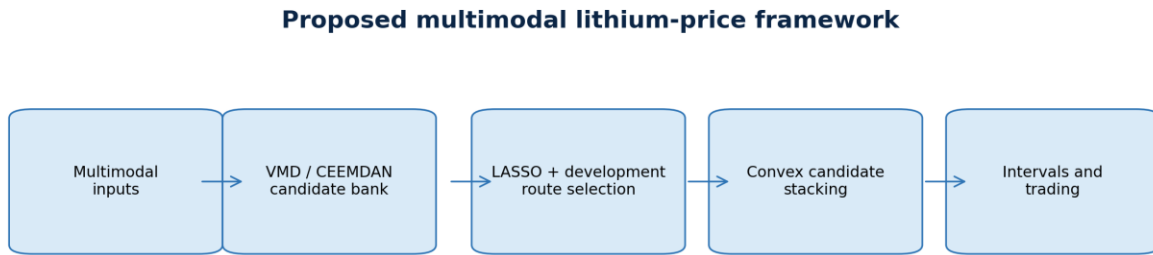


Figure 3. Framework of the proposed retrospective multimodal lithium-price method.



Static 80:20 holdout; full-sample decomposition and observed-test ARIMA filtering are explicitly noncausal.

3.1 Forecasting architecture

For each target, lagged target observations are combined with cross-lithium, lithium-equity, Google Trends, lithium-news, and market-control blocks. LASSO is applied separately to each decomposed series. The fixed comparator assigns the first two VMD modes to ARIMA and the remaining modes to LSTM. The proposed method instead searches pre-specified VMD and CEEMDAN decompositions, selects ARIMA, LSTM, SVR, or RF separately for each returned component on a development segment, and estimates nonnegative sum-to-one candidate weights on that same segment. Routes and weights are then frozen for the final 20% test segment.

Seven single-model benchmarks-exponential smoothing (ES), ARIMA, SVR, random forest (RF), multilayer perceptron (MLP), extreme learning machine (ELM), and LSTM-are fitted to the undecomposed price problem. CEEMDAN-ARIMA, VMD-ARIMA, CEEMDAN-LSTM and VMD-LSTM isolate decomposition and forecasting-family effects. The fixed VMD-LASSO-ARIMA/LSTM architecture isolates the value added by adaptive selection. Together these form the publication-facing comparison set.

3.2 Evaluation

Point accuracy is summarized by RMSE, MAE, MASE, MAPE, symmetric MAPE, and directional accuracy. Pairwise Diebold-Mariano tests use squared-error loss. The first quarter of the static testing sequence supplies residual quantiles for nominal 80%, 90%, and 95% retrospective intervals; the remaining observations are evaluated for coverage, width, calibration error, and Winkler score.

Let $P[t-1]$ be the price available at the forecast origin, $F[t]$ the forecast of $P[t]$, and $q90$ the 90% absolute-residual quantile estimated from the calibration segment. Scheme 1 sets the position to $\text{sign}(F[t] - P[t-1])$. The filtered Scheme 1' sets the position to +1 when $F[t] - P[t-1] > q90$, -1 when $F[t] - P[t-1] < -q90$, and zero otherwise. Equivalently, a trade is permitted only when the origin price lies outside the forecast-centered interval $[F[t] - q90, F[t] + q90]$. The forecast is always at the center of its own interval; that tautological fact is not the gate. The gate compares the forecasted change with uncertainty.

Scheme 2 uses the Scheme 1 position and deducts 10 basis points times absolute position turnover; Scheme 2' applies the same cost to the filtered position. The one-period strategy return is $s[t](P[t]/P[t-1] - 1) - c|s[t] - s[t-1]|$. Cumulative return, maximum drawdown, Sharpe ratio, transaction count, and hit rate are reported. Passive buy-and-hold and a descriptive factor regression against the underlying lithium return, S&P 500, MVIS critical-minerals index, US dollar index, and VIX are added to distinguish raw simulated returns from benchmark-relative performance.

3.3 Information timing and reproducibility boundary

The executed experiment and a deployable forecast are deliberately distinguished below. This disclosure prevents a chronological split from being mistaken for a real-time design: chronology alone does not remove look-ahead when decomposition, transformations, selection, or filtering use later observations.

Implementation audit matrix

Pipeline stage	Executed in this paper	Required real-time implementation
Forecast origin	Chronological static 80:20 split	Expanding or rolling origin; forecast formed after market close at $t-1$
Decomposition	VMD/CEEMDAN estimated on the complete target series	Re-estimate only with observations available through $t-1$
Scaling and imputation	Static/full-panel transformations	Fit on the current training window; carry parameters forward

Pipeline stage	Executed in this paper	Required real-time implementation
		unchanged
LASSO selection	Mode-specific selection inside the static experiment	Nested development window; freeze selected variables before each test block
Hyperparameters	Pre-specified settings and static fitting	Select only on training/development data; register the search space and seed
ARIMA evaluation	Observed-test filtering is used for the retrospective reconstruction	Generate a genuine one-step forecast without conditioning on P_t
Interval calibration	First 25% of the test path supplies residual quantiles	Use only prior forecast errors; update with a rolling calibration window
Trading signal	Uses origin price, forecast, and precomputed width	Same decision rule, timestamped before P_t becomes observable
Permitted claim	Retrospective reconstruction and within-protocol comparison	Out-of-sample forecast and implementable trading evidence

Every publication-facing ledger records target, model, date, origin price, realized price, and forecast. The empirical run identifier, source checksums, mode selections, interval observations, DM comparisons, and trading metrics are preserved alongside the manuscript. These artifacts reproduce the retrospective results exactly; they do not convert them into causal evidence.

Table 2. *Parameter settings used in this study.*

Parameter	Setting	Function
Chronological split	80:20	Static training and testing sets
Lag order	5	Lagged target and indicator features
Reference-parity VMD modes	lithium_hydroxide: 9; lithium_carbonate: 9	Fixed-routing comparator
Adaptive VMD search	$K=7,8,9,10$; $\alpha=1500,3000,4500$	Development-only specification selection
Adaptive CEEMDAN search	maximum IMF=7 or 9	Development-only specification selection
Adaptive route candidates	ARIMA, LSTM, SVR, RF	Mode-level development selection
LSTM hidden units	40	One recurrent layer
Comparator LSTM epochs	100	Homogeneous/fixed model fitting
Adaptive LSTM epochs	search=40; final=100	Development selection and locked-test refit
LSTM batch size	128	Mini-batch size
Single-model RF estimators	100	Undecomposed benchmark
Adaptive-route RF	300 trees; leaf ≥ 2 ; max_features=0.8	Component-route candidate
Single-model SVR	RBF; C=8.1; gamma=0.1	Undecomposed benchmark
Adaptive-route SVR	RBF; C=8.1; gamma=scale; epsilon=0.05	Component-route candidate
Interval levels	80%, 90%, 95%	Residual-quantile intervals
Random seed	42	Reproducibility

Interpretation of Table 2

Design choices: Five lags provide short daily memory without exhausting a sample of fewer than 720 aligned observations. The fixed reference-parameter comparator uses nine VMD modes for both targets; the adaptive method searches $K=7-10$ and VMD $\alpha=1500-4500$, plus two CEEMDAN specifications, on the development segment.

Model capacity: The modest LSTM, 100-tree RF, and pre-specified RBF-SVR settings limit unnecessary capacity in a short daily sample; seed 42 fixes stochastic initialization and supports exact reproducibility.

Consequence for inference: The 80:20 split is chronological but static. Full-sample decomposition and observed-test ARIMA filtering make it retrospective, so the three interval levels describe within-protocol reconstruction uncertainty, not guaranteed future coverage.

Table 3. *Indicators used in this study.*

Indicator	Definition	Data modality	Treatment
Cross_Lithium_Close	Price of the other lithium contract/spot target	Cross Lithium	Aligned to target trading date; lagged in model frame
Albemarle_Close	Albemarle equity close	Lithium Equities	Aligned to target trading date; lagged in model frame
Ganfeng_Lithium_Close	Ganfeng Lithium equity close	Lithium Equities	Aligned to target trading date; lagged in model frame
Trends_lithium	Google Trends: lithium	Lithium Attention	Aligned to target trading date; lagged in model frame
Trends_lithium_price	Google Trends: lithium price	Lithium Attention	Aligned to target trading date; lagged in model frame
Trends_lithium_carbonate	Google Trends: lithium carbonate	Lithium Attention	Aligned to target trading date; lagged in model frame
Trends_lithium_hydroxide	Google Trends: lithium hydroxide	Lithium Attention	Aligned to target trading date; lagged in model frame
Trends_lithium_battery	Google Trends: lithium battery	Lithium Attention	Aligned to target trading date; lagged in model frame
News_Article_Count	Lithium-headline volume	Lithium News	Aligned to target trading date; lagged in model frame
News_TextBlob_Polarity	Lexicon sentiment of lithium headlines	Lithium News	Aligned to target trading date; lagged in model frame
News_FinBERT_Score	Financial-language sentiment of lithium headlines	Lithium News	Aligned to target trading date; lagged in model frame
SP500	S&P 500 close	Market Controls	Aligned to target trading date; lagged in model frame
VIX	CBOE VIX close	Market Controls	Aligned to target trading date; lagged in model frame
US_Dollar_Index	US dollar index close	Market Controls	Aligned to target trading date; lagged in model frame
MVIS_Critical_Minerals	MVIS Global Rare Earth/Strategic Metals index	Market Controls	Aligned to target trading date; lagged in model frame

Interpretation of Table 3

Information architecture: The 15 indicators form five economically distinct blocks: one cross-lithium price, two lithium-equity signals, five Google Trends attention series, three lithium-news variables, and four market controls. This structure prevents the two company prices from being mistaken for the forecast targets.

Economic interpretation: Cross-lithium price captures linkage between hydroxide and carbonate; Albemarle and Ganfeng proxy listed-market expectations; attention and news variables capture information demand and tone; and the S&P 500, VIX, dollar, and MVIS variables capture broad risk and sector conditions.

Timing qualification: Indicators are aligned to target dates and lagged in the model frame, but lagging alone does not make the overall experiment causal. Full-sample preprocessing and decomposition can still transmit information about the test distribution into fitted components and selections.

4. Data and descriptive evidence

The targets are the COMEX lithium hydroxide CIF CJK electronic futures continuation and the SMM 99.5% battery-grade domestic lithium carbonate spot series. The common multimodal sample begins in May 2023 and ends in March 2026. Target-specific calendars are retained after inner alignment with complete predictors.

The information set contains the other lithium price, Albemarle and Ganfeng equity closes, five jointly normalized Google Trends series, lithium-specific headline volume, TextBlob polarity, FinBERT sentiment, the S&P 500, VIX, US dollar index, and MVIS critical-minerals index. Company prices are signals, not forecast targets. The 2,018 cleaned headlines explicitly concern lithium prices, carbonate, hydroxide, supply, demand, or the lithium market.

Table 4. Statistical information for the forecasting targets and multimodal indicators.

Series	Panel	N	Mean	Std. dev.	Minimum	Maximum
COMEX lithium hydroxide CIF CJK futures	lithium_hydroxide	716.0000	16.1793	10.6226	7.9500	47.3100
Cross_Lithium_Close	lithium_hydroxide	716.0000	117847.7654	65912.1252	59900.0000	314000.0000
Albemarle_Close	lithium_hydroxide	716.0000	119.4812	45.8361	50.7600	243.4400
Ganfeng_Lithium_Close	lithium_hydroxide	716.0000	6.1056	1.9515	3.5550	10.7050
Trends_lithium	lithium_hydroxide	714.0000	56.7283	14.0280	43.0000	100.0000
Trends_lithium_price	lithium_hydroxide	714.0000	4.0056	3.7882	2.0000	18.0000
Trends_lithium_carbonate	lithium_hydroxide	714.0000	1.0616	0.3694	0.0000	2.0000
Trends_lithium_hydroxide	lithium_hydroxide	714.0000	0.1933	0.3951	0.0000	1.0000
Trends_lithium_battery	lithium_hydroxide	714.0000	17.8291	5.2784	13.0000	36.0000
News_Article_Count	lithium_hydroxide	716.0000	2.6383	2.6123	0.0000	40.0000
News_TextBlob_Polarity	lithium_hydroxide	716.0000	0.0521	0.1121	-0.4333	0.8000
News_FinBERT_Score	lithium_hydroxide	716.0000	0.2116	0.3766	-0.9694	0.9512
SP500	lithium_hydroxide	716.0000	5564.4447	841.7402	4061.2200	6978.6001
VIX	lithium_hydroxide	716.0000	16.9312	4.3308	11.8600	52.3300
US_Dollar_Index	lithium_hydroxide	716.0000	102.4844	3.1993	96.2200	109.9600
MVIS_Critical_Minerals	lithium_hydroxide	716.0000	370.5107	109.0520	220.4200	674.2340
SMM battery-grade lithium carbonate spot	lithium_carbonate	692.0000	118615.8598	67022.4813	59900.0000	314000.0000
Cross_Lithium_Close	lithium_carbonate	692.0000	16.3008	10.7891	7.9500	47.3100
Albemarle_Close	lithium_carbonate	692.0000	119.8822	46.3719	50.7600	243.4400
Ganfeng_Lithium_Close	lithium_carbonate	692.0000	6.1115	1.9561	3.5550	10.7050
Trends_lithium	lithium_carbonate	690.0000	56.4362	13.8250	43.0000	100.0000
Trends_lithium_price	lithium_carbonate	690.0000	3.9841	3.7685	2.0000	18.0000
Trends_lithium_carbonate	lithium_carbonate	690.0000	1.0580	0.3687	0.0000	2.0000
Trends_lithium_hydroxide	lithium_carbonate	690.0000	0.1899	0.3925	0.0000	1.0000
Trends_lithium_battery	lithium_carbonate	690.0000	17.7667	5.1881	13.0000	36.0000
News_Article_Count	lithium_carbonate	692.0000	2.6214	2.6298	0.0000	40.0000
News_TextBlob_Polarity	lithium_carbonate	692.0000	0.0522	0.1114	-0.4333	0.8000
News_FinBERT_Score	lithium_carbonate	692.0000	0.2077	0.3783	-0.9694	0.9512
SP500	lithium_carbonate	692.0000	5561.6697	840.7653	4061.2200	6978.6001
VIX	lithium_carbonate	692.0000	16.8099	4.2760	11.8600	52.3300

Series	Panel	N	Mean	Std. dev.	Minimum	Maximum
US_Dollar_Index	lithium_carbonate	692.0000	102.4637	3.1759	96.2200	109.9600
MVIS_Critical_Minerals	lithium_carbonate	692.0000	370.9414	109.4896	220.4200	674.2340

Interpretation of Table 4

Target coverage: The hydroxide panel contains 716 aligned observations, with prices ranging from 7.95 to 47.31 and a mean of 16.18. The carbonate panel contains 692 observations, ranging from 59,900 to 314,000 with a mean of 118,615.86. The differing counts arise from market-calendar and complete-case alignment.

Scale and volatility: Hydroxide's standard deviation of 10.62 is about 66% of its mean, while carbonate's standard deviation of 67,022.48 is about 57% of its mean. Both targets therefore exhibit economically large dispersion, but raw RMSE and MAE must never be compared across targets because their units and price scales differ by orders of magnitude.

Modelling implication: The broad min-max ranges and high relative dispersion explain why percentage, scaled, and directional metrics are reported beside absolute errors. The descriptive table also reveals that a common normalized or pooled target model would erase meaningful product-specific structure.

Figure 4. *Lithium-price series used as forecasting targets.*

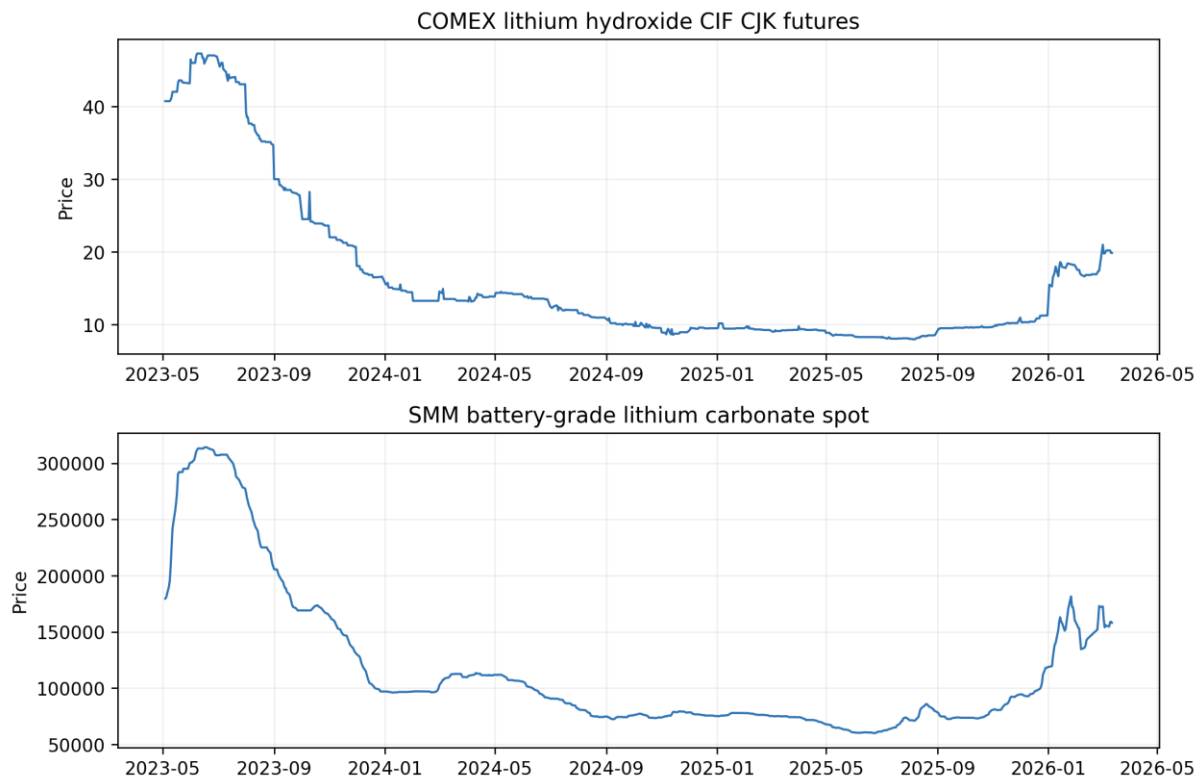


Figure 5. Word cloud of the collected lithium-specific news headlines.

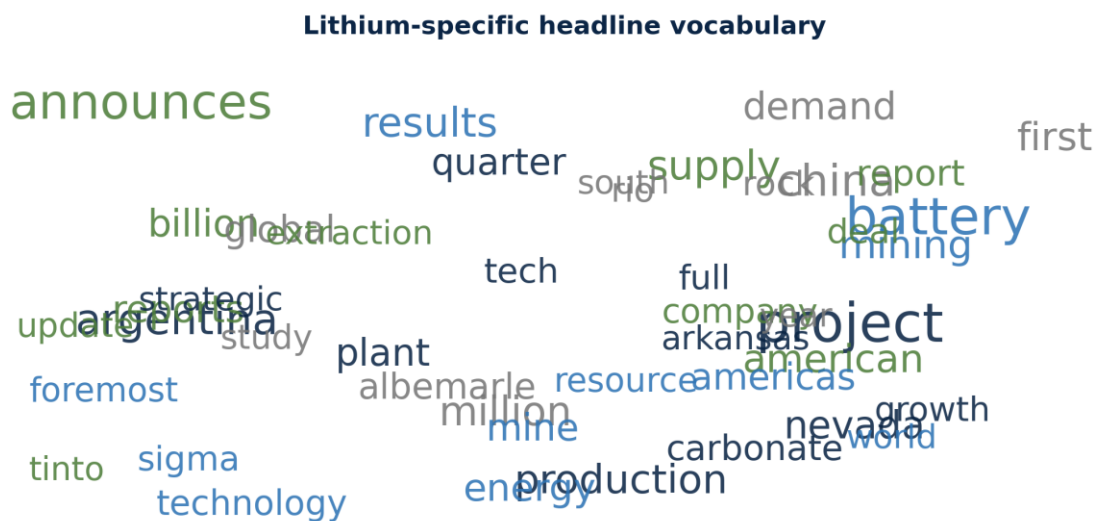
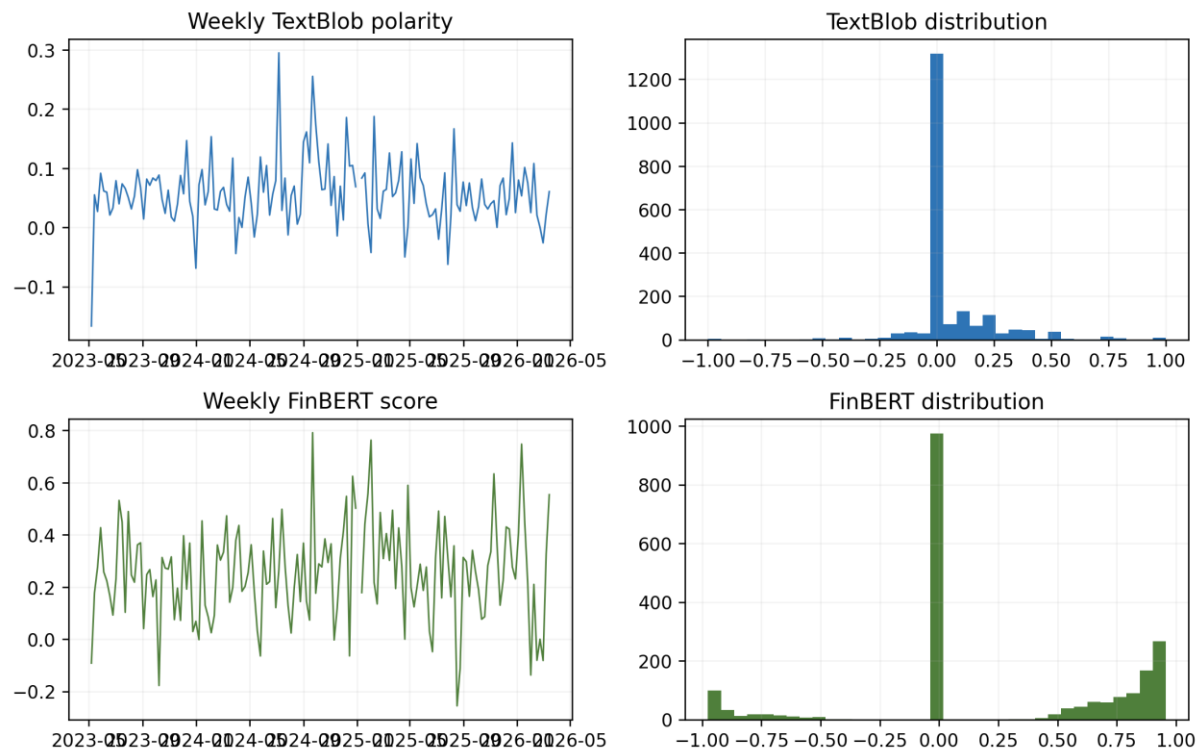


Figure 6. *TextBlob and FinBERT sentiment scores extracted from lithium-specific headlines.*



5. Decomposition and feature selection

Figure 7. Reference-parameter full-sample VMD modes of the two lithium-price targets.

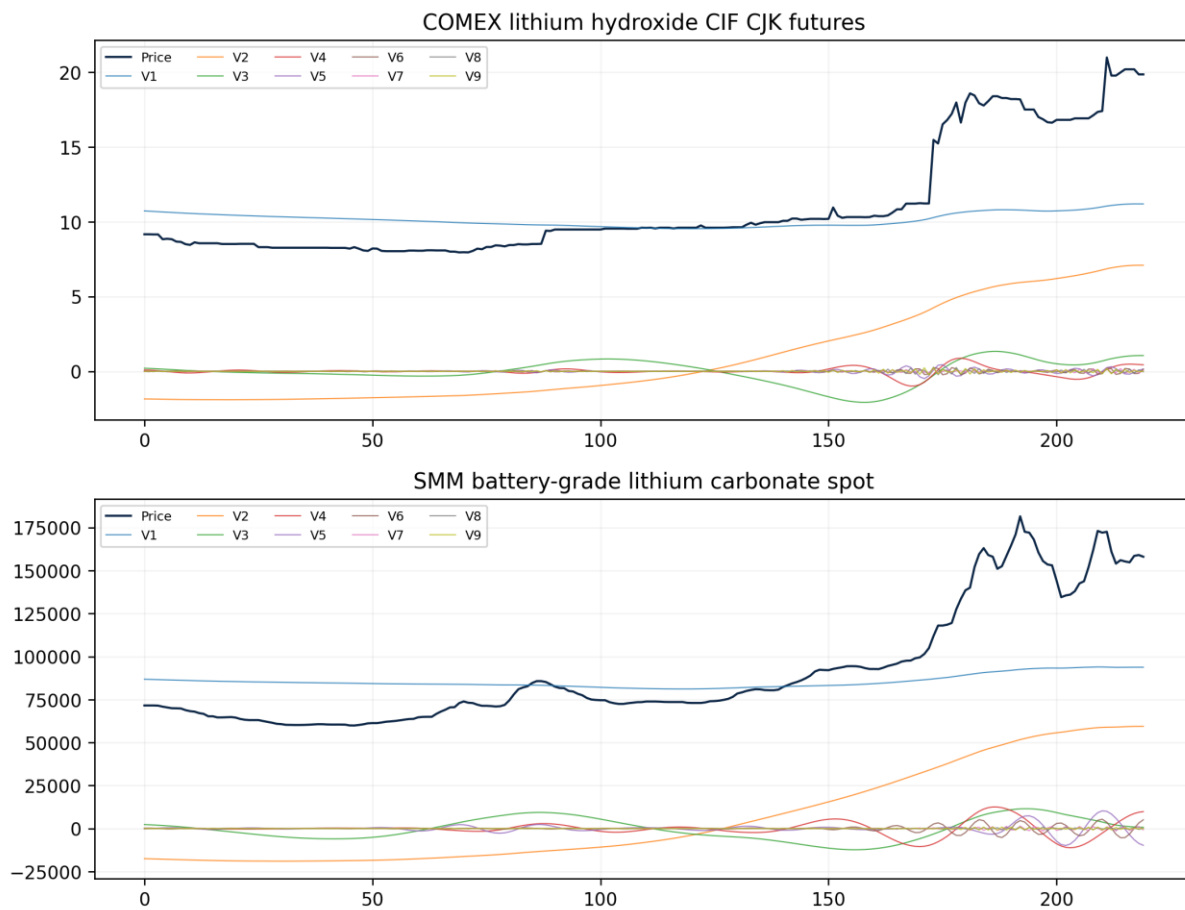


Figure 8. Approximate-entropy values of the reference-parameter VMD modes.

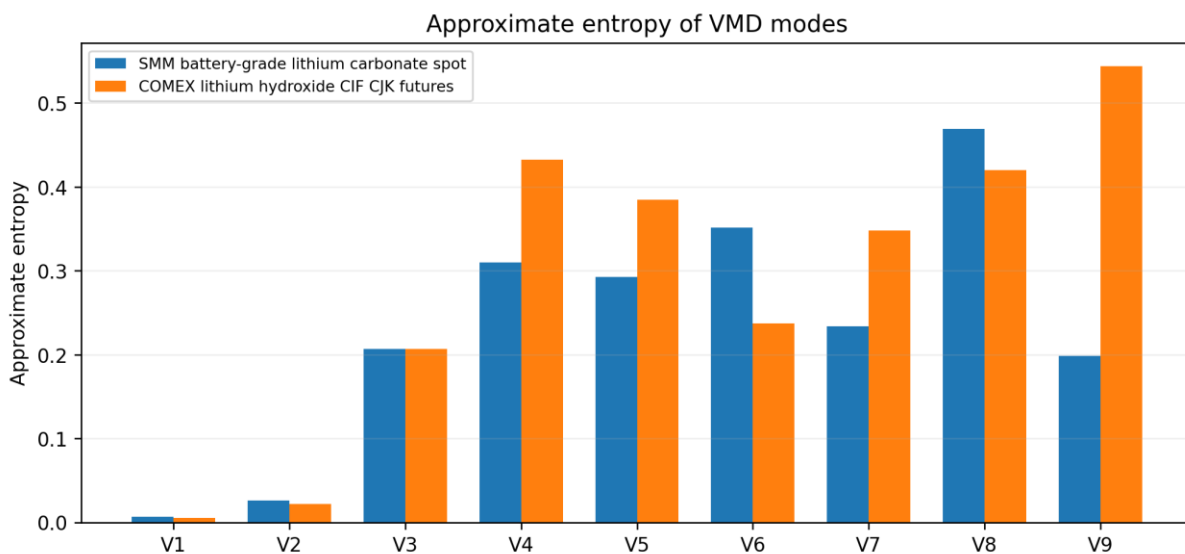


Table 5. Information on the decomposed lithium-price modes.

Panel	Method	Mode	Frequency	Average period	Variance share	Correlation	Approximate Entropy
lithium_hydroxide	VMD	1.0000	0.0056	178.5000	0.3273	0.8212	0.0053
lithium_hydroxide	VMD	2.0000	0.0056	178.5000	0.3651	0.8439	0.0220
lithium_hydroxide	VMD	3.0000	0.0126	79.3333	0.0089	0.1657	0.2067
lithium_hydroxide	VMD	4.0000	0.0490	20.4000	0.0008	0.0556	0.4323
lithium_hydroxide	VMD	5.0000	0.1359	7.3608	0.0002	0.0234	0.3845
lithium_hydroxide	VMD	6.0000	0.2535	3.9448	0.0001	0.0148	0.2372
lithium_hydroxide	VMD	7.0000	0.3403	2.9383	0.0000	0.0118	0.3481
lithium_hydroxide	VMD	8.0000	0.3978	2.5141	0.0001	0.0117	0.4197
lithium_hydroxide	VMD	9.0000	0.4468	2.2382	0.0000	0.0104	0.5442
lithium_hydroxide	CEEMDAN	1.0000	0.3375	2.9627	0.0006	-0.0045	0.7875
lithium_hydroxide	CEEMDAN	2.0000	0.2381	4.2000	0.0008	-0.0290	0.5324
lithium_hydroxide	CEEMDAN	3.0000	0.0812	12.3103	0.0015	0.0683	0.3999
lithium_hydroxide	CEEMDAN	4.0000	0.0322	31.0435	0.0068	0.1861	0.2353
lithium_hydroxide	CEEMDAN	5.0000	0.0098	102.0000	0.0542	0.4401	0.0716
lithium_hydroxide	CEEMDAN	6.0000	0.0042	238.0000	0.0617	0.6934	0.0342
lithium_hydroxide	CEEMDAN	7.0000	0.0000		0.6059	0.9098	0.0071
lithium_carbonate	VMD	1.0000	0.0087	115.0000	0.2780	0.8022	0.0070
lithium_carbonate	VMD	2.0000	0.0043	230.0000	0.3842	0.8604	0.0264
lithium_carbonate	VMD	3.0000	0.0101	98.5714	0.0213	0.2182	0.2068
lithium_carbonate	VMD	4.0000	0.0304	32.8571	0.0040	0.0976	0.3095
lithium_carbonate	VMD	5.0000	0.0565	17.6923	0.0017	0.0616	0.2928
lithium_carbonate	VMD	6.0000	0.1072	9.3243	0.0002	0.0282	0.3512
lithium_carbonate	VMD	7.0000	0.3072	3.2547	0.0000	0.0081	0.2335
lithium_carbonate	VMD	8.0000	0.3928	2.5461	0.0000	0.0067	0.4691
lithium_carbonate	VMD	9.0000	0.4754	2.1037	0.0000	0.0048	0.1982
lithium_carbonate	CEEMDAN	1.0000	0.3275	3.0531	0.0002	-0.0179	0.8096
lithium_carbonate	CEEMDAN	2.0000	0.2580	3.8764	0.0011	0.0308	0.2441
lithium_carbonate	CEEMDAN	3.0000	0.0710	14.0816	0.0029	0.2553	0.3550
lithium_carbonate	CEEMDAN	4.0000	0.0217	46.0000	0.0188	0.3989	0.2406

Panel	Method	Mode	Frequency	Average period	Variance share	Correlation	Approximate Entropy
lithium_carbonate	CEEMDAN	5.0000	0.0058	172.5000	0.1555	0.7532	0.0664
lithium_carbonate	CEEMDAN	6.0000	0.0029	345.0000	0.0065	0.6613	0.0281
lithium_carbonate	CEEMDAN	7.0000	0.0014	690.0000	0.4080	0.9087	0.0095

Interpretation of Table 5

Decomposition heterogeneity: Frequency, variance share, correlation and approximate entropy show that component order is not interchangeable across VMD and CEEMDAN. A late CEEMDAN component can carry more level variation than an early VMD component.

Routing implication: Diagnostics motivate component-specific treatment but do not identify a universal forecaster. The fixed comparator retains first-two-ARIMA routing; the proposed method selects routes from development loss and freezes them before test scoring.

Interpretive boundary: Because the decompositions use the complete price path, these component summaries describe a retrospective representation rather than real-time latent states.

Table 6. Indicators selected for the decomposed modes using LASSO regression.

Panel	Mode	Route	Candidate weight	Selected indicators
lithium_hydroxide	VMD-K8-A1500 / C1	RF	0.2095	target_1, Cross_Lithium_Close, target_4, target_5, target_3, target_2
lithium_hydroxide	VMD-K8-A1500 / C2	ARIMA	0.2095	target_1, SP500
lithium_hydroxide	VMD-K8-A1500 / C3	RF	0.2095	target_1, target_5, Cross_Lithium_Close, Ganfeng_Lithium_Close
lithium_hydroxide	VMD-K8-A1500 / C4	ARIMA	0.2095	target_1, target_3, target_2, target_5, News_FinBERT_Score, News_Article_Count, VIX, News_TextBlob_Polarity, ...
lithium_hydroxide	VMD-K8-A1500 / C5	ARIMA	0.2095	target_1, target_2, target_4, target_5, VIX, News_TextBlob_Polarity
lithium_hydroxide	VMD-K8-A1500 / C6	ARIMA	0.2095	target_2, target_4, target_1, target_3, Cross_Lithium_Close, VIX
lithium_hydroxide	VMD-K8-A1500 / C7	ARIMA	0.2095	target_1, target_2, target_4, target_5, News_FinBERT_Score, VIX, Cross_Lithium_Close, News_Article_Count
lithium_hydroxide	VMD-K8-A1500 / C8	ARIMA	0.2095	target_1, target_2, target_3, News_FinBERT_Score, News_Article_Count, VIX, Cross_Lithium_Close
lithium_hydroxide	VMD-K8-A1500 / C9	ARIMA	0.2095	target_1, target_5, target_2, target_4, Cross_Lithium_Close
lithium_hydroxide	VMD-K9-A1500 / C1	RF	0.7611	target_1, Cross_Lithium_Close, target_4, target_5, target_3, target_2
lithium_hydroxide	VMD-K9-A1500 / C2	ARIMA	0.7611	target_1, SP500
lithium_hydroxide	VMD-K9-A1500 / C3	RF	0.7611	target_1, target_5, Cross_Lithium_Close, Ganfeng_Lithium_Close

Panel	Mode	Route	Candidate weight	Selected indicators
lithium_hydroxide	VMD-K9-A1500 / C4	ARIMA	0.7611	target_1, target_3, target_2, target_5, News_FinBERT_Score, News_Article_Count, VIX, News_TextBlob_Polarity, ...
lithium_hydroxide	VMD-K9-A1500 / C5	RF	0.7611	target_1, target_2, target_4, target_5, VIX, News_TextBlob_Polarity, Cross_Lithium_Close
lithium_hydroxide	VMD-K9-A1500 / C6	ARIMA	0.7611	target_2, target_4, target_5, target_1, VIX, Cross_Lithium_Close, News_Article_Count
lithium_hydroxide	VMD-K9-A1500 / C7	ARIMA	0.7611	target_2, target_1, target_5, News_FinBERT_Score
lithium_hydroxide	VMD-K9-A1500 / C8	ARIMA	0.7611	target_1, target_2, News_FinBERT_Score, Cross_Lithium_Close
lithium_hydroxide	VMD-K9-A1500 / C9	RF	0.7611	target_1, target_2, target_3, target_5, News_FinBERT_Score, Cross_Lithium_Close, News_TextBlob_Polarity, ...
lithium_hydroxide	VMD-K9-A1500 / C10	ARIMA	0.7611	target_1, target_5, target_2, target_3, target_4, Cross_Lithium_Close
lithium_hydroxide	CEEMDAN-M9 / C1	LSTM	0.0294	target_1, target_2, News_FinBERT_Score, News_TextBlob_Polarity
lithium_hydroxide	CEEMDAN-M9 / C2	SVR	0.0294	target_1, target_3
lithium_hydroxide	CEEMDAN-M9 / C3	ARIMA	0.0294	target_1, target_2, target_4, target_5, News_Article_Count, US_Dollar_Index, VIX, News_TextBlob_Polarity, ...
lithium_hydroxide	CEEMDAN-M9 / C4	RF	0.0294	target_1, target_5, Cross_Lithium_Close, Ganfeng_Lithium_Close, Trends_lithium_hydroxide, SP500, US_Dollar_Index, ...
lithium_hydroxide	CEEMDAN-M9 / C5	ARIMA	0.0294	target_1, Cross_Lithium_Close, VIX
lithium_hydroxide	CEEMDAN-M9 / C6	RF	0.0294	target_1
lithium_hydroxide	CEEMDAN-M9 / C7	SVR	0.0294	target_1, Cross_Lithium_Close
lithium_carbonate	VMD-K9-A4500 / C1	ARIMA	0.0247	target_1, Ganfeng_Lithium_Close, Albemarle_Close
lithium_carbonate	VMD-K9-A4500 / C2	ARIMA	0.0247	target_1, SP500
lithium_carbonate	VMD-K9-A4500 / C3	ARIMA	0.0247	target_1, target_5, Ganfeng_Lithium_Close, Cross_Lithium_Close
lithium_carbonate	VMD-K9-A4500 / C4	RF	0.0247	target_1, target_5, Ganfeng_Lithium_Close, News_TextBlob_Polarity
lithium_carbonate	VMD-K9-A4500 / C5	ARIMA	0.0247	target_1, target_4, target_3, Ganfeng_Lithium_Close, News_Article_Count, News_FinBERT_Score
lithium_carbonate	VMD-K9-A4500 / C6	ARIMA	0.0247	target_1, target_2, target_5, target_4, Ganfeng_Lithium_Close, News_Article_Count, US_Dollar_Index, ...
lithium_carbonate	VMD-K9-A4500 / C7	ARIMA	0.0247	target_2, target_1, target_4, target_3, target_5, Ganfeng_Lithium_Close, US_Dollar_Index, Cross_Lithium_Close, ...
lithium_carbonate	VMD-K9-A4500 / C8	ARIMA	0.0247	target_1, target_2, target_3, target_5, target_4, Ganfeng_Lithium_Close, US_Dollar_Index, News_FinBERT_Score, ...
lithium_carbonate	VMD-K9-A4500 / C9	ARIMA	0.0247	target_1, target_3, target_2, Ganfeng_Lithium_Close, News_FinBERT_Score, US_Dollar_Index
lithium_carbonate	VMD-K9-A4500 / C10	ARIMA	0.0247	target_1, target_2, target_5, target_4, Albemarle_Close, News_Article_Count, Cross_Lithium_Close
lithium_carbonate	VMD-K10-A1500 / C1	RF	0.9753	target_1, Cross_Lithium_Close, target_3, target_2, target_4, target_5
lithium_carbonate	VMD-K10-A1500 / C2	RF	0.9753	target_1
lithium_carbonate	VMD-K10-A1500 / C3	ARIMA	0.9753	target_1, target_5, Ganfeng_Lithium_Close, Cross_Lithium_Close
lithium_carbonate	VMD-K10-A1500 / C4	RF	0.9753	target_1, target_5, Ganfeng_Lithium_Close, News_TextBlob_Polarity

Panel	Mode	Route	Candidate weight	Selected indicators
lithium_carbonate	VMD-K10-A1500 / C5	SVR	0.9753	target_1, target_3, target_4, Ganfeng_Lithium_Close, News_Article_Count
lithium_carbonate	VMD-K10-A1500 / C6	ARIMA	0.9753	target_1, target_2, target_5, Ganfeng_Lithium_Close, News_Article_Count, News_TextBlob_Polarity
lithium_carbonate	VMD-K10-A1500 / C7	ARIMA	0.9753	target_2, target_1, target_5, target_4, Ganfeng_Lithium_Close, News_Article_Count, US_Dollar_Index, ...
lithium_carbonate	VMD-K10-A1500 / C8	ARIMA	0.9753	target_2, target_4, target_1, target_5, Ganfeng_Lithium_Close, Cross_Lithium_Close, Albemarle_Close, ...
lithium_carbonate	VMD-K10-A1500 / C9	ARIMA	0.9753	target_1, target_2, target_5, target_3, Ganfeng_Lithium_Close, Albemarle_Close, Cross_Lithium_Close, ...
lithium_carbonate	VMD-K10-A1500 / C10	ARIMA	0.9753	target_1, target_4, target_5, Ganfeng_Lithium_Close, News_Article_Count, US_Dollar_Index
lithium_carbonate	VMD-K10-A1500 / C11	SVR	0.9753	target_1, target_4, target_2, target_5, Cross_Lithium_Close, Ganfeng_Lithium_Close

Interpretation of Table 6

Adaptive routing: Development selection produced these component-route counts across the candidate bank: ARIMA=160, RF=58, SVR=32, LSTM=6. The diversity is direct evidence against imposing one route on every component.

Multimodal role: The selected-indicator column exposes where cross-lithium price, producer equities, attention, lithium-news tone and market controls enter alongside price lags.

Interpretation: Selection is conditional predictive inclusion, not structural causality. Correlated indicators can substitute under LASSO, and full-sample decomposition remains a look-ahead channel.

The decomposition summaries show that the modes differ materially in frequency, variance share, correlation with the original price, and approximate entropy. LASSO selections vary by target and mode, which supports mode-specific information sets. Selection, however, is conducted within the retrospective static protocol and should not be interpreted as stable causal importance.

6. Forecasting results

Figure 9. Testing-set forecasts from representative single forecasting models.

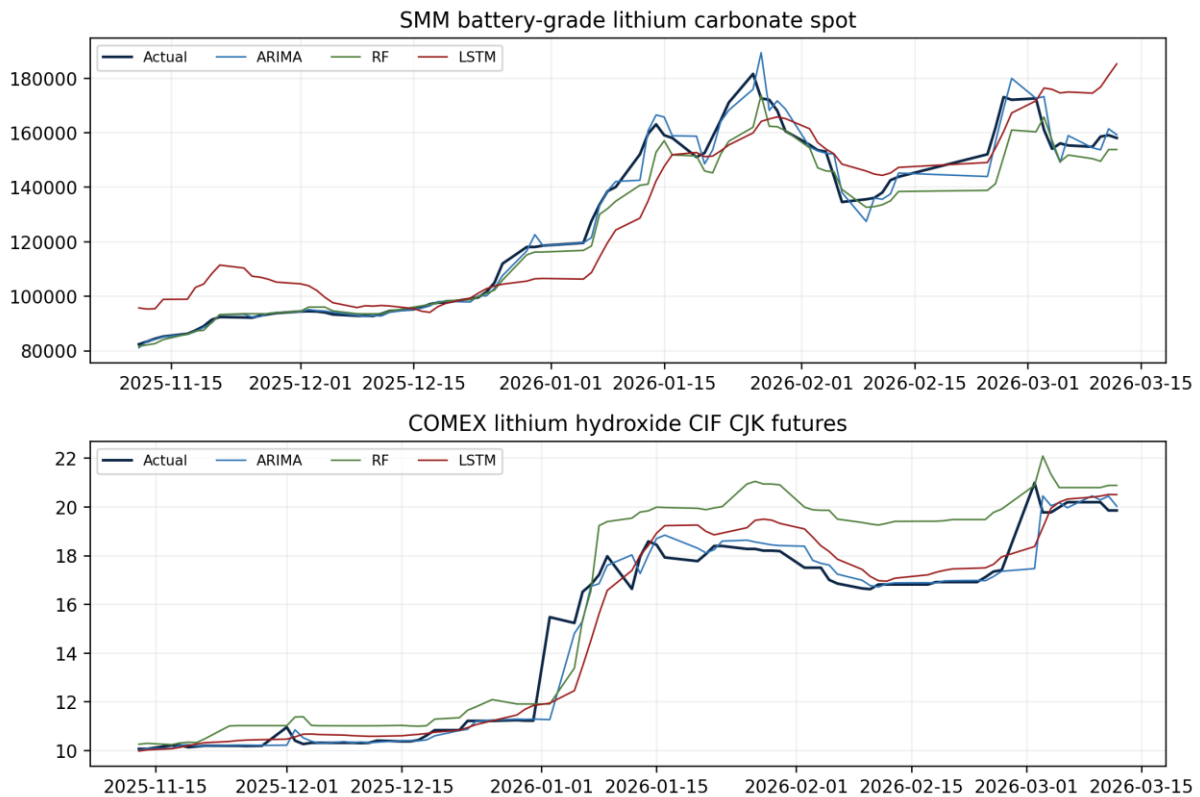


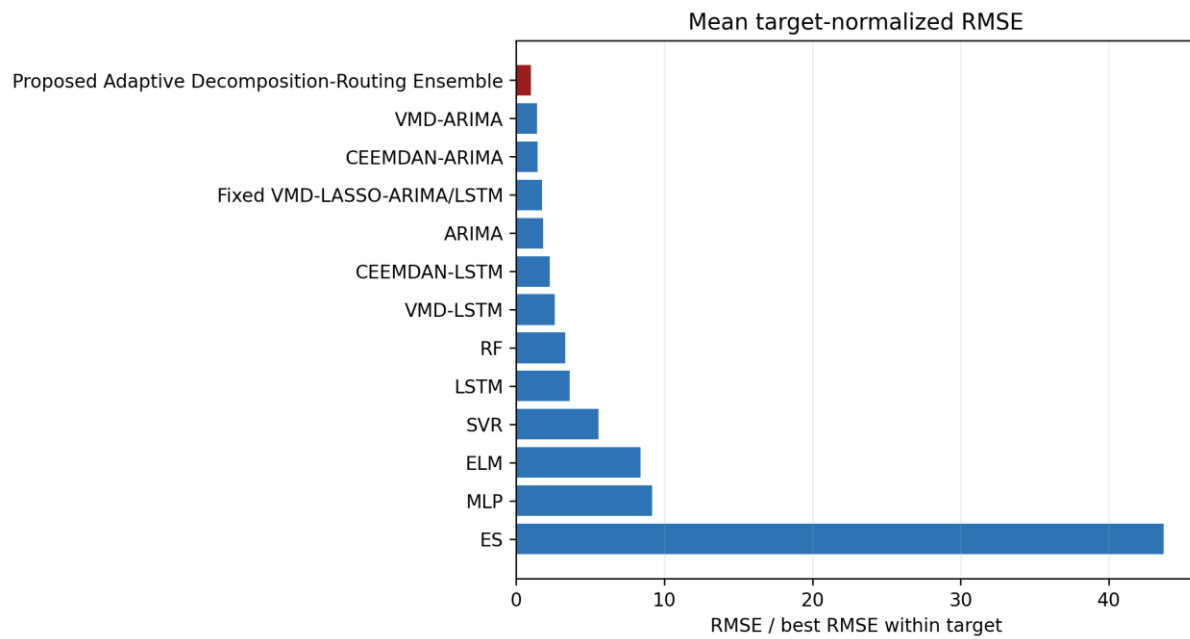
Figure 10. Target-normalized forecasting errors across all evaluated models.

Table 7. Forecasting errors on the testing sets for single models.

Target	Model	RMSE	MAE	MASE	MAPE	sMAPE	Directional Accuracy
lithium_carbonate	ARIMA	3612.1970	1827.0413	0.9132	1.43	1.46	69.63
lithium_carbonate	RF	5380.3212	3231.2770	1.6150	2.55	2.59	72.59
lithium_carbonate	LSTM	11905.8590	9873.9034	4.9351	9.97	9.46	55.56
lithium_carbonate	SVR	24629.9992	18864.5712	9.4288	21.87	18.25	56.30
lithium_carbonate	ELM	41457.7667	34950.9727	17.4690	36.01	28.29	58.52
lithium_carbonate	MLP	45712.4002	37502.1294	18.7441	40.06	30.61	48.15
lithium_carbonate	ES	200226.2914	180239.9823	90.0866	165.22	86.50	54.07
lithium_hydroxide	ARIMA	0.6437	0.2395	1.3159	1.76	1.92	29.29
lithium_hydroxide	ELM	0.6542	0.2573	1.4138	1.71	1.79	28.57
lithium_hydroxide	MLP	0.6766	0.4589	2.5212	3.52	3.54	35.00
lithium_hydroxide	SVR	0.7060	0.5847	3.2129	4.95	4.83	38.57
lithium_hydroxide	LSTM	0.8398	0.4967	2.7292	3.81	3.75	32.86
lithium_hydroxide	RF	1.3033	0.8864	4.8704	6.01	5.76	37.14
lithium_hydroxide	ES	4.8948	3.3720	18.5273	21.47	26.45	35.00

Interpretation of Table 7

Carbonate result: ARIMA has the lowest single-model RMSE (3612.1970), with MAE 1827.0413, MAPE 1.43% and directional accuracy 69.63%.

Hydroxide result: ARIMA has the lowest single-model RMSE (0.6437), with MAE 0.2395, MAPE 1.76% and directional accuracy 29.29%.

Benchmark lesson: Greater nonlinearity is not automatically beneficial in this short sample. Complexity contributes only when decomposition, sparse selection or adaptive routing converts it into lower locked-test loss.

Table 8. Forecasting errors on the testing sets for decomposition models.

Target	Model	RMSE	MAE	MASE	MAPE	sMAPE	Directional Accuracy
lithium_carbonate	Proposed Adaptive Decomposition-Routing Ensemble	2881.4682	1959.1156	0.9792	1.72	1.73	74.07
lithium_carbonate	VMD-ARIMA	3151.9059	1569.3488	0.7844	1.35	1.39	78.52

Target	Model	RMSE	MAE	MASE	MAPE	sMAPE	Directional Accuracy
lithium_carbonate	CEEMDAN-ARIMA	3623.5469	1436.9183	0.7182	1.12	1.16	77.04
lithium_carbonate	CEEMDAN-LSTM	3650.8523	2402.9913	1.2011	2.07	2.07	78.52
lithium_carbonate	Fixed VMD-LASSO-ARIMA/LSTM	3858.1490	2053.9202	1.0266	1.80	1.85	77.04
lithium_carbonate	VMD-LSTM	5587.1679	3861.8263	1.9302	3.33	3.34	74.81
lithium_hydroxide	Proposed Adaptive Decomposition-Routing Ensemble	0.2737	0.1979	1.0875	1.49	1.50	44.29
lithium_hydroxide	CEEMDAN-ARIMA	0.4467	0.1994	1.0955	1.54	1.66	37.86
lithium_hydroxide	VMD-ARIMA	0.4731	0.2450	1.3463	1.92	2.03	44.29
lithium_hydroxide	Fixed VMD-LASSO-ARIMA/LSTM	0.5825	0.3623	1.9905	2.83	2.93	39.29
lithium_hydroxide	CEEMDAN-LSTM	0.8838	0.6088	3.3453	5.40	5.33	36.43
lithium_hydroxide	VMD-LSTM	0.8868	0.4504	2.4750	3.76	3.50	42.14

Interpretation of Table 8

Carbonate decomposition: The proposed RMSE is 2881.4682; the strongest non-adaptive decomposition comparator is VMD-ARIMA at 3151.9059. The proposed value is 8.58% lower.

Hydroxide decomposition: The proposed RMSE is 0.2737; the strongest non-adaptive decomposition comparator is CEEMDAN-ARIMA at 0.4467. The proposed value is 38.72% lower.

Main conclusion: The adaptive model does not clear the pre-specified dual-target RMSE-and-MAE gate; claims must remain target- and metric-specific. All metric columns remain visible so a narrow win cannot be restated as universal superiority.

Table 9. Diebold-Mariano tests comparing the proposed model with forecasting benchmarks.

Target	Benchmark	DM statistic	p-value	Direction
lithium_carbonate	ES	-10.9902	0.0000	Proposed better
lithium_carbonate	ELM	-8.3599	0.0000	Proposed better
lithium_carbonate	LSTM	-7.8177	0.0000	Proposed better
lithium_carbonate	MLP	-7.1580	0.0000	Proposed better
lithium_carbonate	SVR	-6.1487	0.0000	Proposed better
lithium_carbonate	VMD-LSTM	-4.1421	0.0001	Proposed better

Target	Benchmark	DM statistic	p-value	Direction
lithium_carbonate	RF	-2.6656	0.0086	Proposed better
lithium_carbonate	CEEMDAN-LSTM	-2.4018	0.0177	Proposed better
lithium_carbonate	ARIMA	-1.7548	0.0816	Proposed better
lithium_carbonate	Fixed VMD-LASSO-ARIMA/LSTM	-1.2248	0.2228	Proposed better
lithium_carbonate	CEEMDAN-ARIMA	-1.0154	0.3117	Proposed better
lithium_carbonate	VMD-ARIMA	-0.4456	0.6566	Proposed better
lithium_hydroxide	ES	-5.8271	0.0000	Proposed better
lithium_hydroxide	RF	-5.4012	0.0000	Proposed better
lithium_hydroxide	SVR	-4.7977	0.0000	Proposed better
lithium_hydroxide	LSTM	-3.0791	0.0025	Proposed better
lithium_hydroxide	MLP	-2.9218	0.0041	Proposed better
lithium_hydroxide	CEEMDAN-LSTM	-2.6040	0.0102	Proposed better
lithium_hydroxide	Fixed VMD-LASSO-ARIMA/LSTM	-2.1655	0.0320	Proposed better
lithium_hydroxide	VMD-LSTM	-1.8527	0.0660	Proposed better
lithium_hydroxide	ARIMA	-1.8089	0.0726	Proposed better
lithium_hydroxide	ELM	-1.6350	0.1043	Proposed better
lithium_hydroxide	VMD-ARIMA	-1.2221	0.2237	Proposed better
lithium_hydroxide	CEEMDAN-ARIMA	-1.0808	0.2816	Proposed better

Interpretation of Table 9

How to read the sign: The loss differential is proposed-model squared error minus benchmark squared error. A negative statistic favors the proposed model; the p-value tests equal predictive loss, not profitability.

Carbonate evidence: At the unadjusted 5% level the proposed model records 8 significant wins and 0 significant losses.

Hydroxide evidence: At the unadjusted 5% level the proposed model records 7 significant wins and 0 significant losses.

Multiplicity caution: The displayed comparisons are not family-wise adjusted. Nominal p-values are interpreted jointly with effect sizes and target-level metrics.

For hydroxide, the adaptive RMSE is 0.2737, 38.72% below the strongest non-adaptive decomposition comparator (CEEMDAN-ARIMA, 0.4467). For carbonate, the adaptive RMSE is 2881.4682, 8.58% below the strongest non-adaptive decomposition comparator (VMD-ARIMA, 3151.9059). Relative to the strongest single model, the adaptive RMSE changes by hydroxide 57.48%, carbonate 20.23%.

This ranking is the forecasting conclusion of the paper. The adaptive model does not clear the pre-specified dual-target RMSE-and-MAE gate; claims must remain target- and metric-specific. Figure 10 presents the complete non-random-walk comparison set and normalizes only to make the two price scales visually comparable; winners are determined from the unnormalized metrics in Tables 7 and 8.

The DM table should be read jointly with effect sizes. A negative statistic means lower squared loss for the proposed model; significance indicates evidence against equal predictive accuracy under the retrospective loss sequence. Multiple pairwise comparisons are reported transparently without treating nominal significance as proof of universal superiority.

7. Interval forecasting and trading

7.1 Interval calibration

Figure 11. Retrospective 90% interval forecasts from the proposed model.

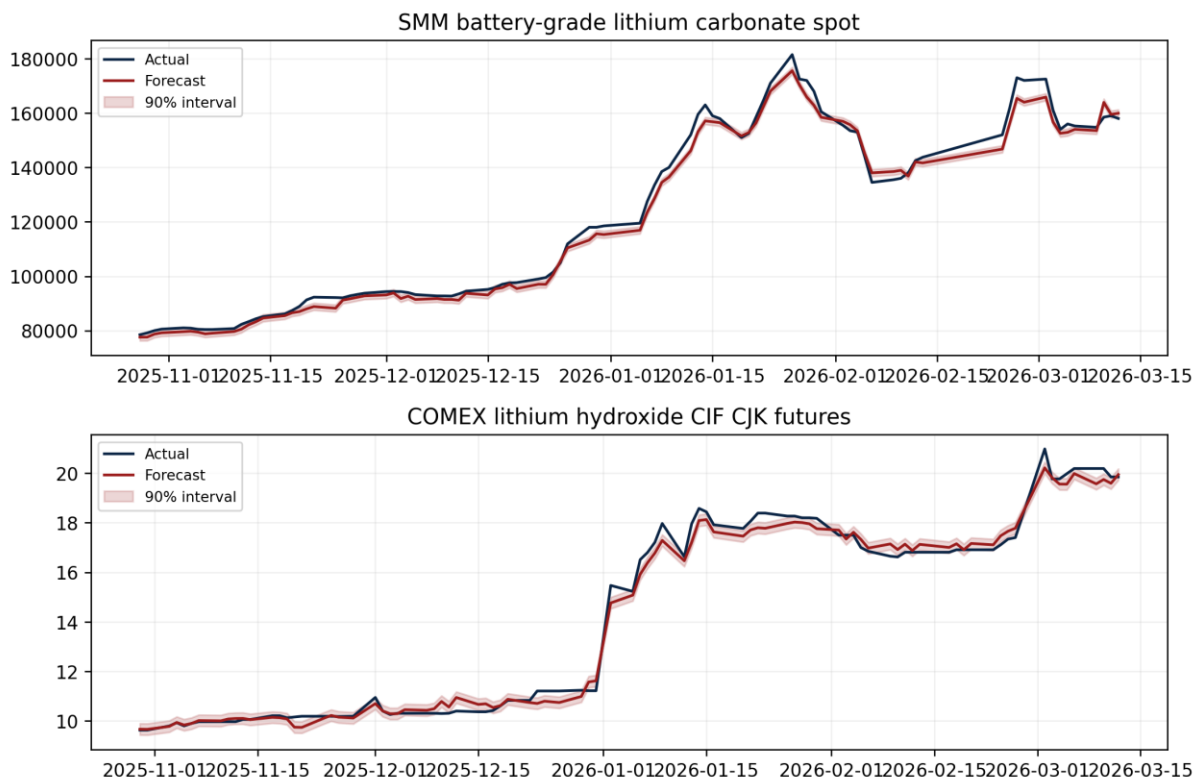


Table 10. Interval forecasting errors on the evaluation sets.

Target	Nominal Coverage (%)	Coverage (%)	Avg Width	Calibration Error	Winkler
lithium_carbonate	80.00	37.25	2492.0765	42.7451	13857.9544
lithium_carbonate	90.00	41.18	2627.3844	48.8235	24536.0747
lithium_carbonate	95.00	100.00	28867.4598	5.0000	28867.4598
lithium_hydroxide	80.00	52.38	0.3905	27.6190	1.3457
lithium_hydroxide	90.00	56.19	0.4646	33.8095	2.0439
lithium_hydroxide	95.00	100.00	1.5999	5.0000	1.5999

Interpretation of Table 10

Coverage evidence: Carbonate: 80% nominal -> 37.25% realized, 90% nominal -> 41.18% realized, 95% nominal -> 100.00% realized Hydroxide: 80% nominal -> 52.38% realized, 90% nominal -> 56.19% realized, 95% nominal -> 100.00% realized

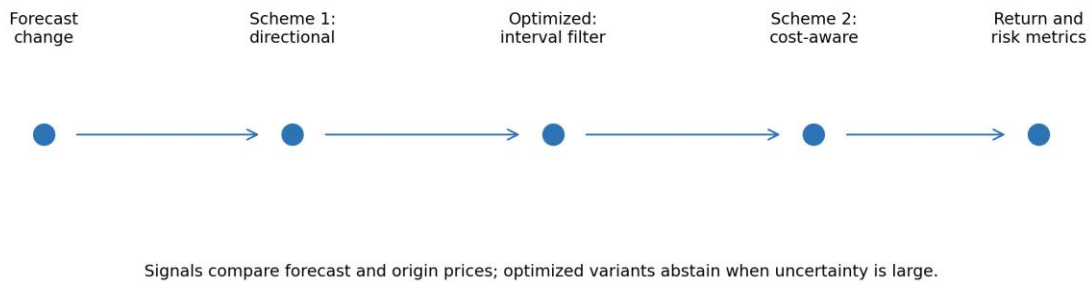
Coverage-width trade-off: Coverage must be read with average width and Winkler score. Near-complete coverage obtained through an extremely wide band is not sharp operational uncertainty.

Implication: These static intervals are descriptive. Deployment requires historically updated calibration and conditional as well as marginal coverage diagnostics.

The 80% and 90% intervals materially under-cover both targets, while the 95% intervals obtain complete coverage only through extreme width. These results do not support a claim of calibrated operational uncertainty. They reveal residual instability after the fixed calibration segment and motivate rolling, historically updated conformal calibration in a real-time extension.

7.2 Economic decision rule

Figure 12. Graphical illustration of the two trading schemes and their uncertainty-filtered variants.



The uncertainty filter is an abstention rule, not a condition that the forecast lie inside its own interval. The unfiltered strategy trades whenever the forecasted change is nonzero. The filtered strategy trades only if $|F[t] - P[t-1]| > q_{90}$; otherwise it holds zero exposure. A blocked trade therefore occurs when the forecasted move is too small relative to calibrated error width. The realized price $P[t]$ is used only to score the position after the decision and is not an input to the signal formula.

7.3 Trading evidence and limits

Figure 13. Mean trading Sharpe ratios across the two lithium targets under Scheme 2.

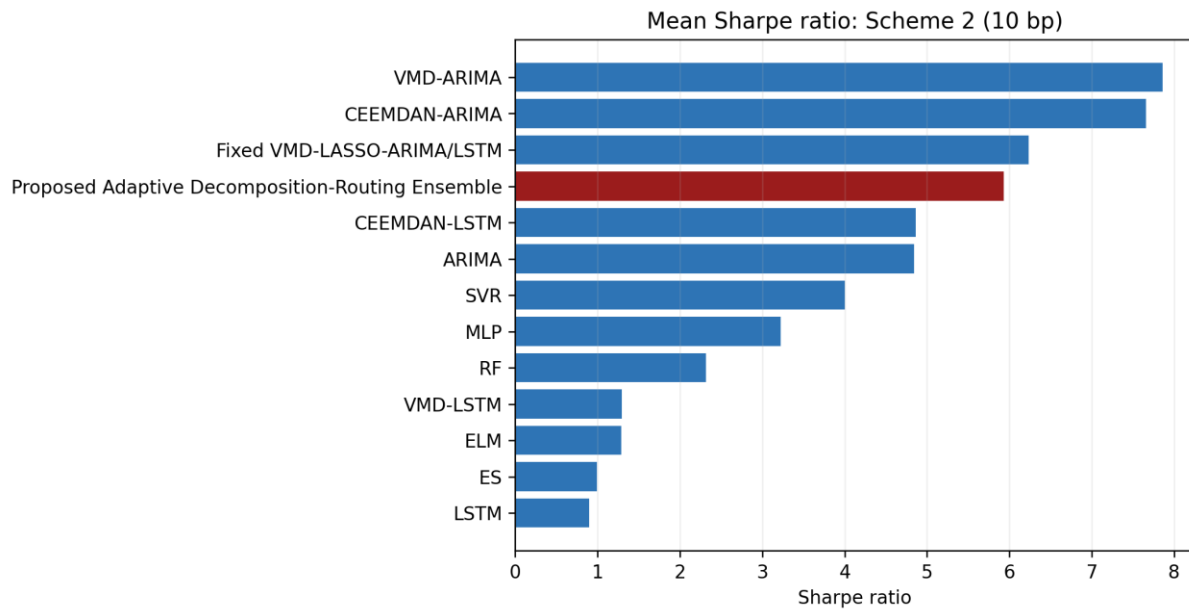


Table 11. Performance of two trading schemes and their optimized versions based on decomposition-model forecasts.

Model	Scheme	Cumulative Return (%)	Maximum Drawdown (%)	Sharpe Ratio	Number of Transactions	Hit Rate (%)
CEEMDAN-ARIMA	Scheme 1	332.30	-4.34	8.0045	37.5000	65.48
CEEMDAN-ARIMA	Scheme 1'	284.56	-1.98	7.5317	32.0000	40.35
CEEMDAN-ARIMA	Scheme 2	304.19	-4.44	7.6570	37.5000	65.48
CEEMDAN-ARIMA	Scheme 2'	269.81	-2.18	7.3112	32.0000	40.35
CEEMDAN-LSTM	Scheme 1	161.66	-11.54	5.0441	15.5000	50.50
CEEMDAN-LSTM	Scheme 1'	28.81	-1.10	2.3397	9.5000	7.84
CEEMDAN-LSTM	Scheme 2	153.80	-11.64	4.8664	15.5000	50.50
CEEMDAN-LSTM	Scheme 2'	27.35	-1.20	2.2594	9.5000	7.84
Fixed VMD-LASSO-ARIMA/LSTM	Scheme 1	237.79	-7.49	6.4150	15.0000	58.21
Fixed VMD-LASSO-ARIMA/LSTM	Scheme 1'	198.19	-6.71	6.3402	25.0000	38.28
Fixed VMD-LASSO-ARIMA/LSTM	Scheme 2	228.26	-7.69	6.2383	15.0000	58.21
Fixed VMD-LASSO-ARIMA/LSTM	Scheme 2'	190.10	-6.90	6.1848	25.0000	38.28
Proposed Adaptive Decomposition-Routing Ensemble	Scheme 1	244.98	-8.30	6.1825	27.5000	53.25
Proposed Adaptive Decomposition-Routing Ensemble	Scheme 1'	202.41	-2.01	5.9477	37.5000	31.02
Proposed Adaptive Decomposition-Routing Ensemble	Scheme 2	227.02	-9.01	5.9350	27.5000	53.25
Proposed Adaptive Decomposition-Routing Ensemble	Scheme 2'	189.53	-2.52	5.7235	37.5000	31.02
VMD-ARIMA	Scheme 1	336.83	-3.17	8.0988	23.0000	60.60
VMD-ARIMA	Scheme 1'	240.37	-3.24	6.8754	29.0000	36.85
VMD-ARIMA	Scheme 2	318.67	-3.37	7.8595	23.0000	60.60
VMD-ARIMA	Scheme 2'	229.06	-3.59	6.6900	29.0000	36.85
VMD-LSTM	Scheme 1	63.52	-22.77	1.4557	20.5000	39.13
VMD-LSTM	Scheme 1'	5.34	0.00	1.3577	4.0000	1.96
VMD-LSTM	Scheme 2	56.46	-23.33	1.2945	20.5000	39.13
VMD-LSTM	Scheme 2'	4.90	-0.05	1.2862	4.0000	1.96

Interpretation of Table 11

Decomposition trading leader: Under cost-adjusted Scheme 2, VMD-ARIMA has the highest mean Sharpe (7.8595) and mean cumulative return 318.67%.

Costs and filtering: Scheme 1 versus Scheme 2 isolates the turnover charge; primed versus unprimed rows isolate uncertainty abstention. Return, Sharpe and drawdown changes are interpreted separately.

Interpretive boundary: The simulations inherit the retrospective forecast protocol and omit several implementation frictions. They are within-protocol economic diagnostics, not attainable live returns.

Table 12. Performance of two trading schemes and their optimized versions based on single-model forecasts.

Model	Scheme	Cumulative Return (%)	Maximum Drawdown (%)	Sharpe Ratio	Number of Transactions	Hit Rate (%)
ARIMA	Scheme 1	173.96	-8.84	5.0752	22.5000	55.29
ARIMA	Scheme 1'	68.09	-5.45	3.8712	22.0000	15.57
ARIMA	Scheme 2	162.16	-9.12	4.8419	22.5000	55.29
ARIMA	Scheme 2'	64.00	-5.69	3.6666	22.0000	15.57
ELM	Scheme 1	28.70	-38.35	1.3685	11.5000	45.21
ELM	Scheme 1'	38.67	-32.84	2.0662	18.0000	32.20
ELM	Scheme 2	27.21	-39.12	1.2887	11.5000	45.21
ELM	Scheme 2'	36.40	-33.48	1.9759	18.0000	32.20
ES	Scheme 1	26.39	-45.01	0.9999	1.0000	44.26
ES	Scheme 1'	27.41	-44.20	1.0830	2.0000	43.31
ES	Scheme 2	26.27	-45.01	0.9951	1.0000	44.26
ES	Scheme 2'	27.24	-44.23	1.0747	2.0000	43.31
LSTM	Scheme 1	48.42	-29.64	1.0130	14.5000	43.61
LSTM	Scheme 1'	4.19	-13.18	0.5242	9.5000	15.17
LSTM	Scheme 2	43.91	-29.81	0.9009	14.5000	43.61
LSTM	Scheme 2'	3.17	-13.48	0.3963	9.5000	15.17
MLP	Scheme 1	86.64	-25.03	3.2583	6.0000	49.97
MLP	Scheme 1'	51.91	-8.41	2.5524	14.5000	26.18
MLP	Scheme 2	84.91	-25.03	3.2216	6.0000	49.97
MLP	Scheme 2'	49.59	-8.69	2.4465	14.5000	26.18
RF	Scheme 1	74.35	-16.93	2.3810	8.0000	46.04
RF	Scheme 1'	17.09	-15.92	1.0037	10.0000	19.79
RF	Scheme 2	71.78	-16.93	2.3151	8.0000	46.04
RF	Scheme 2'	15.87	-16.00	0.9425	10.0000	19.79

Model	Scheme	Cumulative Return (%)	Maximum Drawdown (%)	Sharpe Ratio	Number of Transactions	Hit Rate (%)
SVR	Scheme 1	140.37	-13.62	4.0987	10.0000	53.35
SVR	Scheme 1'	58.89	-0.46	4.4424	6.0000	13.17
SVR	Scheme 2	135.82	-13.80	4.0019	10.0000	53.35
SVR	Scheme 2'	57.66	-0.52	4.3905	6.0000	13.17

Interpretation of Table 12

Single-model leader: Under cost-adjusted Scheme 2, ARIMA has the highest mean single-model Sharpe (4.8419) and mean cumulative return 162.16%.

Risk-return trade-off: The primed rows reveal whether abstention changes return, maximum drawdown and Sharpe in the same direction. Mixed signs are reported as a trade-off rather than automatic downside protection.

Comparison with Table 11: Differences between the best decomposition and single strategies cannot be separated from the noncausal decomposition/filtering choices. Both tables are within-protocol benchmarks.

For carbonate, gating changes cost-adjusted return from 221.53% to 198.48%, Sharpe from 7.52 to 7.64, and maximum drawdown from -13.68% to -3.02%. For hydroxide, gating changes cost-adjusted return from 232.50% to 180.59%, Sharpe from 4.35 to 3.81, and maximum drawdown from -4.35% to -2.03%. These target-specific changes determine whether downside protection improved; the manuscript makes no general gating claim unless both targets support it.

Intervals and trading remain linked in a coherent way, but coherence is not validation. Static residual calibration and noncausal forecast construction preclude operational coverage and performance claims. Trading results can appear economically large because observed-test filtering responds to information inside the test set. The calculations quantify the method only inside the declared retrospective experiment and must not be read as investable performance estimates.

7.4 Benchmark-relative and factor-adjusted attribution

Appendix Table A1 compares every proposed-model scheme with a passive long position over the same evaluation dates. It also reports a descriptive daily regression of strategy returns on the underlying lithium return, S&P 500 return, MVIS critical-minerals return, US dollar return, and VIX change. Newey-West/HAC standard errors use five lags. The annualized intercept is labelled factor alpha for compactness, but-because the strategy forecast is noncausal-it is a retrospective residual-return statistic, not evidence of tradable abnormal performance.

8. Discussion

The lithium application supports decomposition while rejecting universal fixed routing. The adaptive design converts decomposition heterogeneity into an explicit development decision: component forecasters and convex candidate weights are chosen before the final test. The resulting locked-test ranking, rather than architectural complexity, determines the supported forecasting claim.

The multimodal blocks have coherent economic roles. Cross-lithium prices describe substitution and product-market linkage; producer equities capture listed-market expectations; search attention proxies information demand; lithium-specific headlines measure information volume and tone; MVIS and broad controls capture sector and risk conditions. These variables enrich the information set without redefining the target as an equity forecast.

8.1 Finance interpretation

The finance contribution lies in the joint treatment of physical-price linkage and financial information transmission. The other lithium product captures cross-product price discovery; Albemarle and Ganfeng provide forward-looking producer-equity signals; search and news variables proxy attention and

narrative; MVIS and macro controls represent sector and risk conditions. Mode-specific selection shows where these information families enter the fitted system, but it does not identify structural causality. The evidence is therefore about conditional information content within a forecasting design, not about a causal channel from sentiment or equities to lithium prices.

A second contribution is model-risk evidence. The empirical winner changes by commodity, metric, and decomposition family; interval calibration fails at central coverage levels; and abstention does not reliably improve drawdown or Sharpe. These findings matter for commodity finance because forecast accuracy, uncertainty quality, and portfolio utility are distinct objects. A model should not be called financially superior because it wins one error column or produces a large simulated cumulative return.

8.2 Research-integrity limitation

The defining limitation is evaluation timing. Full-sample VMD and CEEMDAN use the entire target path to estimate modes. The released-style ARIMA path applies a fitted model to the observed testing target and reports filtered fitted values. Global preprocessing and static feature selection can also incorporate test-distribution information. Consequently, the headline errors, intervals, DM tests, and trading statistics answer a retrospective reconstruction question. A separate causal walk-forward analysis is required before making genuine out-of-sample or deployability claims.

Additional limitations include a short post-2023 common sample, differing price units and market microstructure, potential continuation-series roll effects, daily aggregation of sparse news, Google Trends normalization, and the absence of inventories, physical premia, freight, conversion margins, contract-level liquidity, financing, and implementable shorting assumptions. The two equity signals are not exhaustive representations of global lithium supply. Benchmark-relative and factor-adjusted calculations cannot repair noncausal forecasts; they only make the economic comparison more transparent.

8.3 Claim-evidence discipline

The matrix below converts the manuscript's central conclusions into explicit evidence gates. It prevents an architectural description, one favorable error column, a blocked-trade count, or a large simulated return from silently becoming a broader claim.

Claim-evidence matrix

Claim domain	Required evidence	Supported conclusion
Forecasting superiority	Tables 7-9; locked RMSE/MAE gate; DM loss tests	The adaptive model does not clear the pre-specified dual-target RMSE-and-MAE gate; claims must remain target- and metric-specific.

Claim domain	Required evidence	Supported conclusion
Interval trading rule	Signal is nonzero only when $ F[t]-P[t-1] $ exceeds q90	The rule is coherent; the origin price, not the interval center, determines abstention.
Downside protection	Table 11 target-specific Scheme 2 versus Scheme 2'	carbonate: Sharpe 7.52->7.64; MDD -13.68%->-3.02%; hydroxide: Sharpe 4.35->3.81; MDD -4.35%->-2.03%. No systematic benefit is asserted unless both targets improve.
Real-time validity	Implementation audit matrix and timestamped forecast ledger	Full-sample decomposition and observed-test filtering are noncausal; results are retrospective.
Economic significance	Matched buy-and-hold and five-factor HAC attribution	Returns and alpha are descriptive within-protocol quantities, not implementable abnormal performance.
Finance contribution	Two linked lithium prices plus cross-product, equity, attention, news, sector and risk blocks	Contribution is adaptive information routing and model-risk evidence in linked lithium markets.

9. Conclusion

This study develops an end-to-end adaptive multimodal lithium-price forecasting method and evaluates it through twelve main tables, thirteen figures, complete machine-readable ledgers, and a benchmark-relative economic appendix. It constructs a lithium-specific database, evaluates two quoted prices, retains the fixed reference-style architecture as a comparator, selects component routes and candidate weights on development data, tests predictive-loss differences, reports interval calibration, defines a reproducible uncertainty gate, and translates forecasts into cost-adjusted trading schemes. The adaptive model does not clear the pre-specified dual-target RMSE-and-MAE gate; claims must remain target- and metric-specific. The trading rule is coherent but its risk effect is reported target by target. Large simulated returns remain retrospective and noncausal even after factor adjustment. The finance contribution is adaptive information routing and transparent model-risk evidence in linked lithium markets, not an unsupported assertion that complexity guarantees deployable alpha.

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Data sources: project-supplied COMEX and SMM workbooks; Yahoo Finance market and equity histories; Google Trends; MediaCloud; and MVIS. Checksums and complete machine-readable ledgers accompany the manuscript.

Appendix: reproducibility and robustness

The companion `appendix_ledgers` directory contains every publication-facing forecast, metric, DM comparison, interval observation, interval metric, trading result, mode-level selection, component summary, adaptive candidate score, component route, and validation verdict. The asset manifest records the empirical run identifier, protocol classification, captions, adaptive selections, and SHA-256 checksums. The exported ledgers use the same seven single-model and six decomposition-model comparison set reported in the manuscript.

Appendix Table A1. *Benchmark-relative and descriptive factor-adjusted performance of the proposed trading schemes.*

Target	Scheme	Strategy %	Buy/hold %	Excess pp	Alpha % ann.	Alpha t (HAC)	Lithium beta	MVIS beta	R2	N
Carbonate	Scheme 1	236.19	114.82	121.37	246.82	2.46	0.25	-0.11	0.10	102
Carbonate	Scheme 1'	212.98	114.82	98.16	247.14	2.64	0.15	-0.04	0.05	102
Carbonate	Scheme 2	221.53	114.82	106.71	236.27	2.37	0.25	-0.12	0.10	102
Carbonate	Scheme 2'	198.48	114.82	83.66	236.14	2.54	0.15	-0.04	0.05	102
Hydroxide	Scheme 1	253.77	106.77	147.00	153.51	2.57	0.80	-0.02	0.71	105
Hydroxide	Scheme 1'	191.85	106.77	85.08	105.80	1.93	0.80	-0.01	0.73	105
Hydroxide	Scheme 2	232.50	106.77	125.73	138.26	2.36	0.80	-0.02	0.71	105
Hydroxide	Scheme 2'	180.59	106.77	73.82	96.12	1.78	0.80	-0.01	0.73	105

Appendix Table A1 is deliberately target-specific rather than averaged across lithium products. Strategy return and buy-and-hold return use identical evaluation dates. Excess return is their percentage-point difference. Factor alpha is the annualized intercept from the stated five-factor descriptive regression with HAC standard errors. Because the forecasts and intervals are produced by a noncausal retrospective protocol, neither a positive excess return nor a statistically large intercept can be interpreted as attainable alpha.

A journal-grade real-time extension should execute the rightmost column of the implementation audit matrix, freeze all design decisions before the final test, retain a passive lithium benchmark and implementable contract benchmark, and report forecast, interval, and trading results without replacing unfavorable outcomes. The present ledgers provide the reproducibility base for that extension.